

Calculus – Group Discussion

Week 6 – Multivariable functions and Partial derivatives

ADSAI & CS / PR12

Learning goals (Week 6)

- Interpret functions of two variables and their domains.
- Understand level curves and surfaces.
- Compute partial derivatives with respect to different variables.
- Compute higher-order and mixed partial derivatives.
- Interpret the gradient vector and its geometric meaning.

Level curves, domains, and graphs

A **function of two variables** assigns a unique real number $f(x, y)$ to each point (x, y) in some subset $D \subseteq \mathbb{R}^2$ called the domain of f .

$$z = f(x, y)$$

is a surface of the graph in $\mathbb{R}^3$.

A **level curve** is obtained by fixing the output:

$$f(x, y) = c.$$

So level curves are 2D slices of the surface at height $z = c$.

Visualization: <https://www.geogebra.org/m/M2P4KsRe>

Partial derivatives

Differentiation can be extended from single-variable functions to multi-variable functions through **partial differentiation**:

$$\frac{\partial}{\partial x} f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h} \quad (\text{partial derivative of } f \text{ w.r.t. } x)$$

$$\frac{\partial}{\partial y} f(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h} \quad (\text{partial derivative of } f \text{ w.r.t. } y)$$

Intuition: vary one of the variables while keeping all other variables fixed.

Slicing visualization: <https://www.desmos.com/calculator/y0h5kuvmbt?lang=nl>

Notations for first partial derivative:

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} f(x, y) = f_1(x, y) = f_x(x, y).$$

Higher-order partial derivatives and mixed derivatives

Once first partial derivatives are known, we can differentiate again:

$$\frac{\partial}{\partial x} \frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial x} \frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial x^2} = f_{xx}(x, y) = f_{11}(x, y)$$

$$\frac{\partial}{\partial y} \frac{\partial}{\partial y} f(x, y) = \frac{\partial}{\partial y} \frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial y^2} = f_{yy}(x, y) = f_{22}(x, y)$$

We can also differentiate in different orders to obtain **mixed partials**:

$$\frac{\partial}{\partial x} \frac{\partial}{\partial y} f(x, y) = \frac{\partial}{\partial x} \frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial x \partial y} = f_{yx}(x, y) = f_{21}(x, y)$$

$$\frac{\partial}{\partial y} \frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial y} \frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial y \partial x} = f_{xy}(x, y) = f_{12}(x, y)$$

If the function is **sufficiently smooth**, then usually $f_{xy} = f_{yx}$.

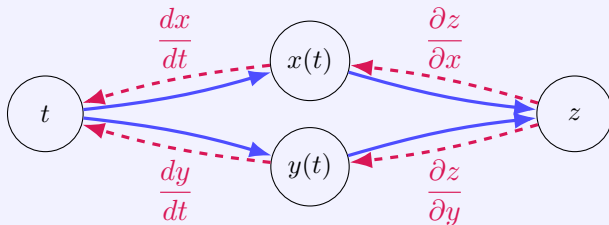
Total chain rule

Chain rule revisited. If

$$z = f(x, y), \quad x = x(t), \quad y = y(t),$$

then $z = f(x(t), y(t))$ and depends on t through both intermediate functions. In this case, there are two paths that lead to t from z :

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$



Gradient vector and directional derivatives

The **gradient** of $f(x, y)$ is the vector

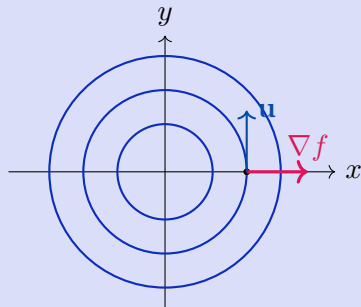
$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial}{\partial x} f(x, y) \\ \frac{\partial}{\partial y} f(x, y) \end{pmatrix} \in \mathbb{R}^2.$$

The gradient is the collection of first partial derivatives.

Geometric meaning. At a point, ∇f points in the direction of **steepest increase**. It is also perpendicular to the level curve through that point.

Visualization:

<https://www.desmos.com/calculator/zqrsm0enu9?lang=nl>



Gradient follows the unit basis vectors $\mathbf{i}, \mathbf{j}$

$$\nabla f = \mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y}$$

and is normal to a level curve
vector $\mathbf{u}$

$$\nabla f \cdot \mathbf{u} = 0$$

Key takeaways

- A function of two variables has a domain in $\mathbb{R}^2$ and a graph in $\mathbb{R}^3$.
- Level curves show where the function takes constant values.
- Partial derivatives measure change in one coordinate direction while the other variable is held fixed.
- Higher-order and mixed partials describe curvature and interacting rates of change.
- The gradient points in the direction of steepest increase and links naturally to directional derivatives.